**Math 526, F09**
**Quiz 10**

**Name:**
**SSN: xxx-xx-**

**<u>Instructions:</u> There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.**

**Problem 1.** (10 points) True or False.

(a) The determinant of $\mathbf{I} + \mathbf{A}$ is $1 + det\mathbf{A}$.

False

(b) The determinant of $\mathbf{ABC}$ is $|\mathbf{A}||\mathbf{B}||\mathbf{C}|$.

True

(c) The determinant of $4\mathbf{A}$ is $4|\mathbf{A}|$.

False

(d) $det(\frac{1}{2}\mathbf{A}) = -\frac{1}{16}$ if $\mathbf{A}$ is a $4 \times 4$ matrix with $det\mathbf{A} = -1$.

True

(e) The determinant of $\mathbf{AB} - \mathbf{BA}$ is zero.

True

**Problem 2.** (10 points) Find the determinant of $\mathbf{A}$ where $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 4 & 5 \\ 5 & 4 & 0 & 3 \\ 2 & 0 & 0 & 1 \end{bmatrix}$ using the co-factor expansion.

$$\det A = 4\cdot(-1)^{2+3}\begin{vmatrix} 1 & 0 & 2 \\ 5 & 4 & 3 \\ 2 & 0 & 1 \end{vmatrix}$$

$$= -4\left[1\cdot(-1)^{1+1}\begin{vmatrix} 4 & 3 \\ 0 & 1 \end{vmatrix} + 2\cdot(-1)^{1+3}\begin{vmatrix} 5 & 4 \\ 2 & 0 \end{vmatrix}\right]$$

$$= -4\left(4+(-16)\right)$$

$$= -4\cdot(-12)$$

$$= 48$$